Ziyad Abouelenin

Contact through https://ziyadabouelenin.github.io/

SUMMARY

Robotics engineer with a Ph.D. in Mechanical and Aerospace Engineering and experience developing robotic systems from modeling and actuator characterization through real-time control and experimental validation. Specializes in manipulator dynamics, system identification, motion and force control, and hardware-software integration. Developed and validated a reusable 4-DoF robotic platform operating at up to 500 Hz, with additional experience in industrial automation, additive manufacturing, sensing, and composite fabrication.
U.S. Citizen.

EDUCATION

Rutgers University **NJ, USA**
Ph.D. in Mechanical and Aerospace Engineering **October 2026**

Dissertation: A Force-Sensitive Robotic Framework for Automated Bandage Peeling.

Graduate Certificate in Socially Cognizant Robotics.

German University in Cairo **Egypt**
B.S. in Mechatronics Engineering **July 2019**

Thesis completed at NYU Tandon School of Engineering: Prosthetic Hand Design and Control Using EMG Signals.

ROBOTICS AND ENGINEERING EXPERIENCE

Rutgers University **NJ, USA**
NSF-NRT SOCRATES Fellow & Researcher **September 2021 - August 2026**

- Developed and experimentally validated a reusable 4-DoF robotic platform for real-time motion control and force-sensitive manipulation, integrating synchronized actuator communication, sensing, and MATLAB/Simulink control.
- Characterized actuator torque, frequency response, hysteresis, and drivetrain friction; identified a fourth-order actuator model with a 34.0 rad/s bandwidth.
- Derived the manipulator kinematics, analytical Jacobian, and symbolic Euler-Lagrange dynamics; estimated physically consistent parameters through constrained optimization.
- Validated the inverse-dynamics model on independent trajectories, achieving $R^2 = 0.93$ for the gravity-dominated shoulder joint.
- Developed and deployed motion and computed-torque controllers at 500 Hz; formulated task-space impedance and passivity-based controllers for force-sensitive peeling.
- Conducted load-cell-based peel experiments and measured a 3.4-fold increase in adhesive fracture energy, from 37 to 126 J/m², across peel rates of 1 to 6.3 mm/s.
- Mentored more than five graduate and undergraduate researchers in robotic experimentation, additive manufacturing, and mechatronic system development.

NASA Marshall Space Flight Center **AL, USA**
Structural Design and Analysis Intern **January 2022 - August 2022**

- Fabricated carbon-fiber–silicone composite specimens through molding and additive manufacturing.
- Characterized their viscoelastic properties using dynamic mechanical analysis, demonstrating that fiber orientation could be tailored to increase stiffness and damping along the loading direction.
- Fitted numerical material models to the experimental data and simulated the structural response of composite vibration-isolation components.

Panacea Technologies Inc. **PA, USA**

Automation Validation Engineer **September 2020 - August 2021**

- Developed a Python application that automated Ignition HMI configuration and reduced development time by 95%.
- Programmed and validated Rockwell control systems using ladder logic, function block diagrams, and sequential function charts.
- Integrated Studio 5000, FactoryTalk, WIN-911, ThinManager, and SQL Server in virtualized industrial-automation environments.
- Executed system testing and validation for integrated PLC, HMI, alarm, and data-management functions.

Stevens Institute of Technology **NJ, USA**

Robotics Researcher **January 2020 - July 2020**

- Developed a CAD and Gazebo model of a Crazyflie 2.1 micro-aerial vehicle.
- Modeled the geometric and physical properties of the vehicle in a Gazebo simulation Environment.
- Implemented geometric control of the simulated vehicle.

NYU Tandon School of Engineering **NY, USA**

Robotics Researcher **February 2018 - July 2018**

- Designed and built a 500 g underactuated, tendon-driven prosthetic hand with four embedded servomotors for grasping, pinching, pointing, and resting gestures.
- Developed a Python pipeline for eight-channel surface EMG acquisition, Gaussian and RMS filtering, and machine-learning-based gesture classification, achieving up to 97% accuracy in 10-fold cross-validation.
- Integrated an SVM classifier with Arduino-based motor control, enabling real-time prosthetic-hand operation with up to 89.6% gesture-specific accuracy and a 0.4 s actuation delay.

SELECTED PUBLICATIONS & PRESENTATIONS

- Ghalyan, I. F., Abouelenin, Z. M., and Kapila, V. “Gaussian Filtering of EMG Signals for Improved Hand Gesture Classification.” 2018 IEEE Signal Processing in Medicine and Biology Symposium, 2018.
- Ghalyan, I. F. J., Abouelenin, Z. M., Annamalai, G., and Kapila, V. “Gaussian Smoothing Filter for Improved EMG Signal Modeling.” Signal Processing in Medicine and Biology, Springer, 2020, pp. 161–204.
- DermAssist: Towards Robotic Wound Care Management, Rutgers Robotics Workshop, 2025.
- Construction, Analysis, and Verification of a 3D-Printed Carbon Fiber-Silicone Composite System for Vibration Isolation, IMECE, 2023.
- Tunable 3D-Printed Damper for In-Space Additive Manufacturing, IMECE, 2022.

SKILLS

Programming & Simulation: MATLAB, Simulink, Python, Simscape Multibody

Robotics & Control: Manipulator dynamics, system identification, computed-torque control, impedance control, passivity-based control, force control, forward/inverse kinematics, Jacobian-based control

Experimental Systems: Real-time control, data acquisition, load-cell sensing, actuator characterization, frequency-response testing, dynamic mechanical analysis

Automation & Manufacturing: Rockwell Studio 5000, FactoryTalk, Ignition, DeltaV, Wonderware, ladder logic, function block diagrams, additive manufacturing, composite fabrication

SELECTED CERTIFICATIONS

- Ignition 8.1 Gold Certified
- Ignition 8.1 Core Certified
- ThinManager Certified Integrator
- Rockwell Certified Integrator